Scalability & Future Plan

Date	26 October 2026
Team ID	Gyanvi Jain
Project Name	Formula 1 Race Intelligence Dashboard
Maximum Marks	4 Marks

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases,

increased data loads, or extended features in the future. It also captures the team's roadmap for

enhancing and evolving the project beyond its current state.

Current System Limitations

S.No.	Limitation	Impact	Priority to Address
1	Dashboard depends on the available Excel/CSV datasets and Tableau Public.	New data must be prepared and the dashboard updated to reflect additional races and seasons.	High
2	Dashboard performance may decrease as data volume and visualisation complexity increase.	Loading and filtering may become slower with larger datasets.	Medium
3	Tableau Public provides limited control over access and deployment.	Sensitive data and restricted user access would require a different hosting approach.	Medium

Scalability Plan

S.No.	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Dashboard published on Tableau Public for browser-based access.	Consider an appropriate hosted analytics platform if user traffic increases significantly.
2	Data Storage	Excel/CSV files and Tableau data sources or extracts.	Migrate to a structured database such as PostgreSQL if the dataset grows substantially.
3	Performance	Tableau visualisations and calculated fields process the available dataset.	Optimise calculations, reduce unnecessary data, and use suitable extracts or database queries.
4	Security	Dashboard is publicly accessible through Tableau Public.	Use controlled access and appropriate authentication if a private deployment is required.

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Add new race seasons and improve data refresh procedures.	Short-term	Keeps the dashboard updated with additional data.
Phase 2	Integrate a structured database and optimise dashboard performance.	Medium-term	Supports larger datasets and more efficient analysis.
Phase 3	Develop an optional web application with enhanced access controls and advanced analytics.	Long-term	Improves accessibility, extensibility, and support for future features.